

Introduction

Boolean algebra is widely used in digital electronics to simplify logic expressions and reduce the number of logic gates required for implementation. In this experiment, four Boolean expressions were simplified using Boolean algebra laws. The original and simplified circuits were implemented and tested to verify that both produce identical outputs.

Task 1: Simplification of Boolean Expression

Overview

The objective of this task is to simplify the given Boolean expression using Boolean algebra and verify the result through logic circuit implementation.

Given Expression

$$F(A, B, C) = A'BC + AB'C + ABC' + ABC$$

Simplification

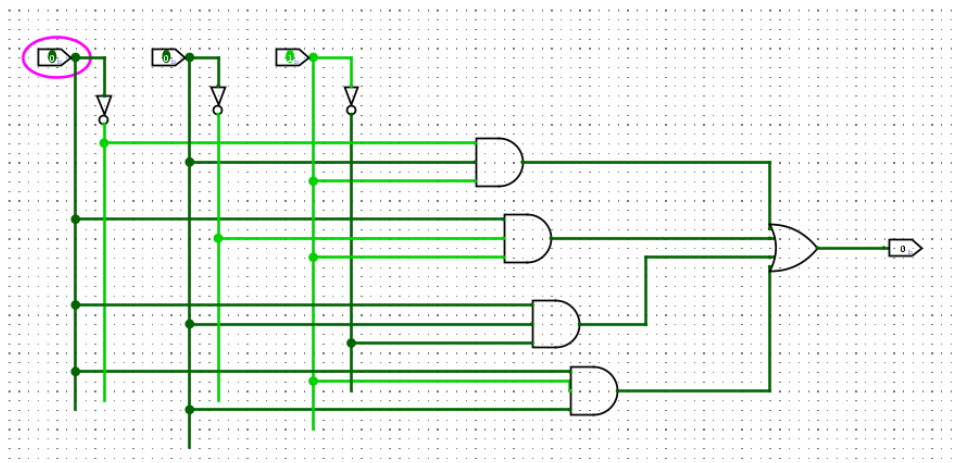
$$F(A, B, C) = A'BC + AB'C + ABC' + ABC$$

$$= A'BC + ABC + AB'C + ABC'$$

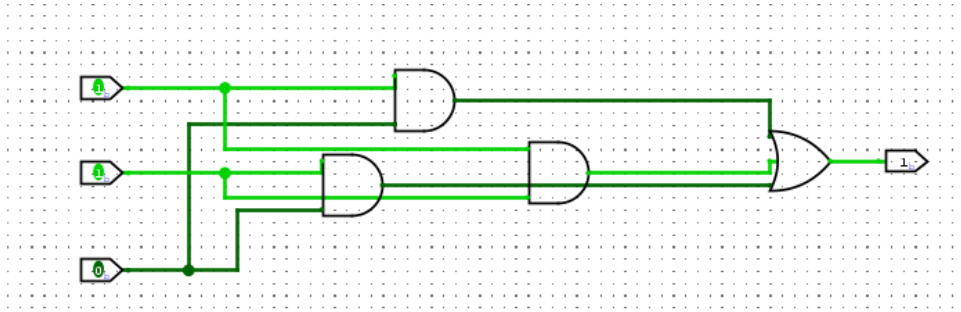
$$= BC + AC + AB$$

$$= AB + AC + BC$$

Original Logic Circuit



Simplified Logic Circuit



Testing

The original and simplified circuits were simulated using all possible combinations of inputs A, B, and C. The outputs were compared and found to be identical.

Conclusion

The original Boolean expression was successfully simplified to

$$F(A, B, C) = AB + AC + BC$$

The simplified circuit requires fewer logic gates while maintaining the same functionality.

Task 2: Simplification of Boolean Expression

Overview

The purpose of this task is to simplify a Boolean expression containing multiple sum terms and verify the result using logic circuits.

Given Expression

$$F(A, B, C) = A(A + B)(A + B + C)$$

Simplification

$$F(A, B, C) = A(A + B)(A + B + C)$$

$$= (A + AB)(A + B + C)$$

$$= A + AB + AC + AB + AB + ABC$$

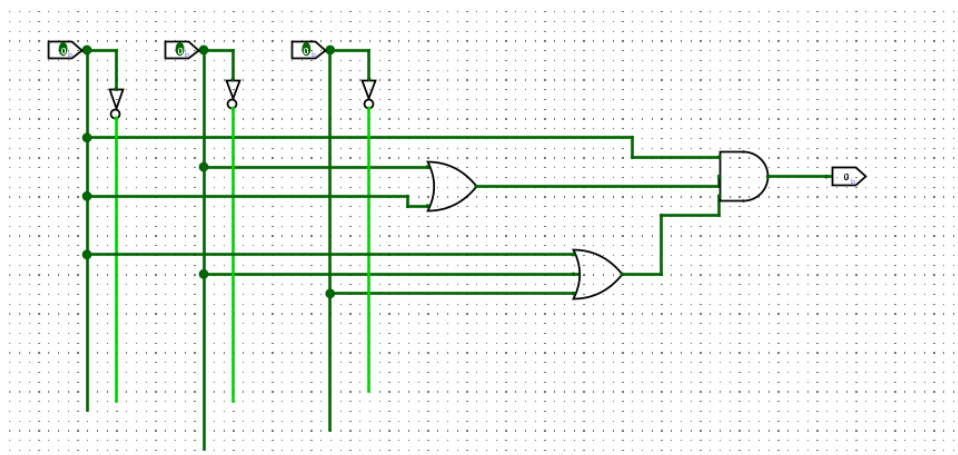
$$= A + AB + AC + ABC$$

$$= A + AB + AC$$

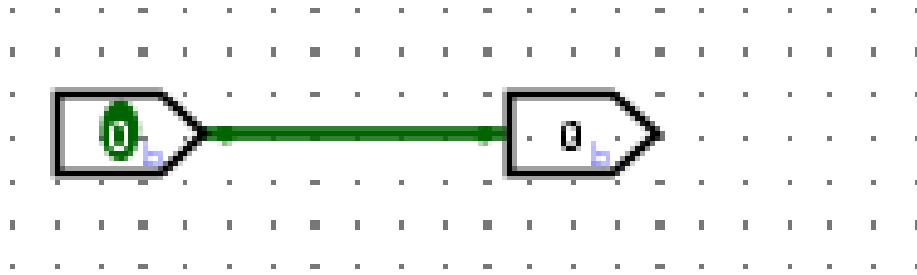
$$= A(1 + B + C)$$

$$= A$$

Original Logic Circuit



Simplified Logic Circuit



Testing

The outputs of the original and simplified circuits were checked for all possible input combinations. Both circuits produced exactly the same outputs.

Conclusion

The expression was simplified to

$$F(A, B, C) = A$$

which significantly reduces circuit complexity.

Task 3: Simplification of Boolean Expression

Overview

This task demonstrates the simplification of a Boolean expression that reduces to a constant logic value.

Given Expression

$$F(A, B, C) = (A + (BC)')(AB + ABC')$$

Simplification

$$F(A, B, C) = (A + (BC)')(AB + ABC')$$

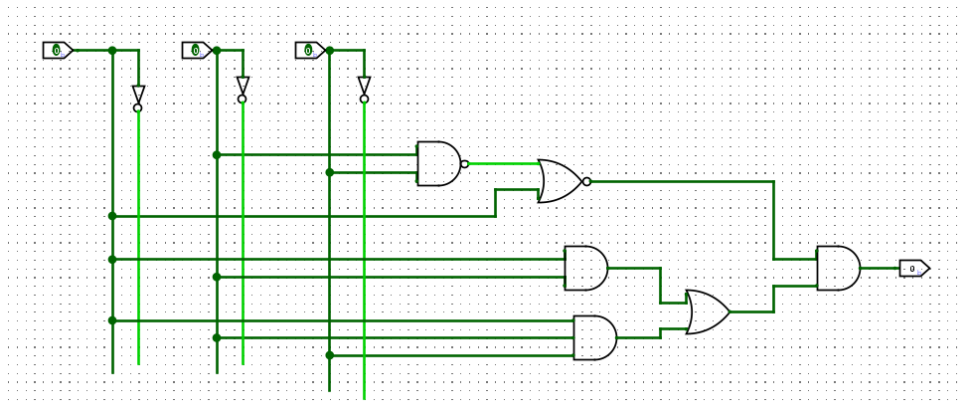
$$= AB(A + (BC)')$$

$$= AB(A + B' + C')$$

$$= AB(A'BC)$$

$$= 0$$

Original Logic Circuit



Testing

The output remained LOW (0) for every possible input combination. The original and simplified circuits produced identical results.

Conclusion

The Boolean expression simplifies to

$$F(A, B, C) = 0$$

Therefore, the output is always logic LOW regardless of the inputs.

Task 4: Simplification of Boolean Expression

Overview

The objective of this task is to simplify the given Boolean expression and verify its equivalence using logic circuit implementation.

Given Expression

$$F(A, B, C) = (B'(A + B) + (A + B)(A + B'))B'$$

Simplification

$$F(A, B, C) = (B'(A + B) + (A + B)(A + B'))B'$$

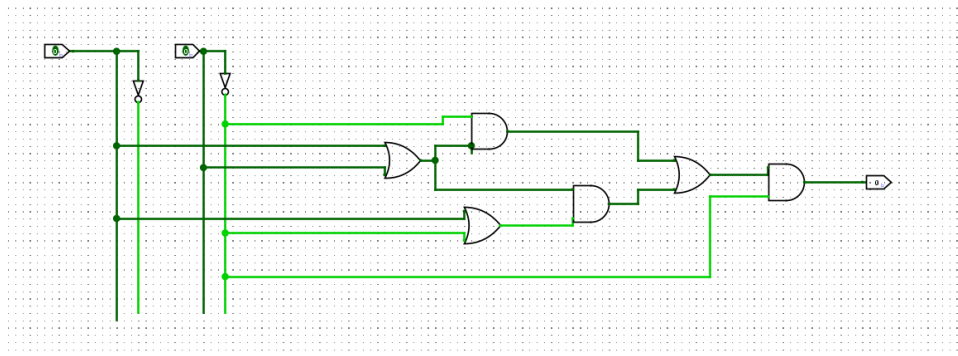
$$= (AB' + A + AB' + AB)B'$$

$$= (A + AB + AB')B'$$

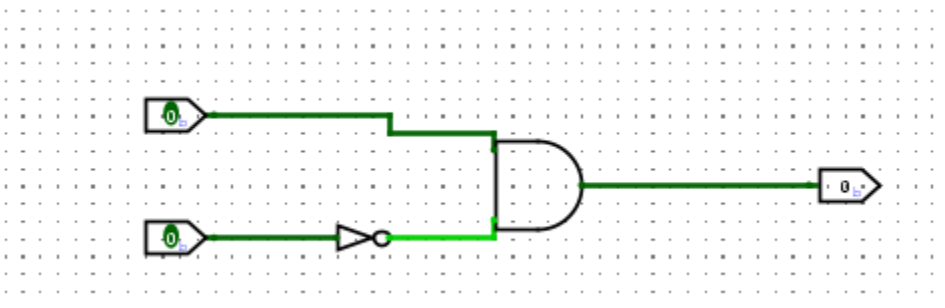
$$= AB'(1 + B + B')$$

$$= AB'$$

Original Logic Circuit



Simplified Logic Circuit



Testing

The outputs of the original and simplified circuits were compared for all possible input combinations. Both circuits produced identical outputs.

Conclusion

The given expression simplifies to

$$F(A, B, C) = AB'$$

The simplified circuit requires fewer gates and provides the same output as the original circuit.

Overall Conclusion

In this experiment, four Boolean expressions were simplified using Boolean algebra laws. The original and simplified logic circuits were implemented and tested. The results confirmed that the simplified expressions are logically equivalent to the original expressions. Boolean simplification reduces hardware requirements, decreases circuit complexity, and improves the efficiency of digital systems.